

1.)

$$0.005 \text{ m}^3$$

$$1.5 \text{ m}^3$$

$$P_1 = 5 \text{ bar}$$

$$v_g = 1.5752 \times 10^{-3}$$

$$v_g = 0.2503$$

$$y = \frac{v_g}{v_g + v_f} = 0.9937$$

$$v = v_g + y(v_f - v_g) = 0.2519 \text{ m}^3/\text{kg}$$

$$v_1 = v_g + y(v_f - v_g) = 0.249 \text{ m}^3/\text{kg}$$

$$v_2 = v_1$$

$$(5 - 5.25)$$

$$v_2 = v_g = 0.249 \text{ m}^3/\text{kg}$$

$$P_2 = 5 + \left(\frac{0.249 - 0.2503}{0.2390 - 0.2503} \right) (5.25 - 5)$$

$$P_2 = 5.029 \text{ bar}$$

$$u_2 = v_g = 1321.02 + \left(\frac{0.249 - 0.2503}{0.2390 - 0.2503} \right) (1322.07 - 1321.02)$$

$$v_2 = 1321.14$$

$$w = 0.424 (0.249 - 1321.14) = -560.06 \text{ kJ}$$

2.)

$$18 \text{ kg}$$

$$100 \text{ kPa}$$

$$-10^\circ \text{C}$$

$$550 \text{ kPa}$$

$$P_1 = 100 \text{ kPa}$$

$$(100) (1.5834)^{1.1} = 550 (v_2)^{1.1}$$

$$\frac{100}{550} = \left(\frac{v_2}{1.5834} \right)^{1.1} \quad v_2 = 0.288 \text{ m}^3/\text{kg}$$

$$w = 18 \left(\frac{100 \cdot 1.5834 - 550 \cdot 0.288}{1.1 - 1} \right)$$

$$w = -10.8 \text{ kJ}$$

$$P_1 = 100 \text{ kPa} @ -10^\circ \text{C}$$

$$h_1 = 1450.54 \text{ kJ/kg}$$

$$P_2 = 550 \text{ kPa} @ v_2 = 0.288 \text{ m}^3/\text{kg}$$

$$h_2 = 1581.38 \text{ kJ/kg}$$

$$Q = 18 (-1450.54 - 1581.38)$$

$$Q = -3031.92 \text{ kJ}$$

$$3) \quad m = 0.25 \text{ kg} \quad p_1 = 2 \text{ bar} \quad T_1 = 200^\circ\text{C} \quad \kappa = 1.4$$

$$200 \text{ kPa} \quad 473.15 \text{ K}$$

$$c_p = 1.005 \text{ kJ/kg}\cdot\text{K} \quad c_v = 0.718 \text{ kJ/kg}\cdot\text{K}$$

$$R_{\text{air}} = 0.287 \text{ kJ/kg}\cdot\text{K}$$

$$V_1 = \frac{0.25 \cdot 0.287 \cdot 10^3}{200 \times 10^3} \cdot 473.15$$

$$V_1 = 0.1697 \text{ m}^3$$

$$V_2 = 2V_1 = 0.339 \text{ m}^3$$

$$T_2 = 2T_1 = 400 \text{ K}$$

$$W = p(V_2 - V_1) = 200 \cdot 10^3 (0.339 - 0.1697)$$

$$W = 33.86 \text{ kJ}$$

$$Q = 0.25 \cdot 1.005 (400 - 473.15)$$

$$Q = -18.88 \text{ kJ}$$

4.)

$$p_1 = 0.7 \text{ bar} \quad T_1 = 275 \text{ K} \quad V_1 = 0.25 \text{ m}^3 \quad V_2 = 0.03 \text{ m}^3$$

$$70 \text{ kPa}$$

$$pV^{1.25}$$

$$n = 1.25 \quad \gamma_{\text{air}} = 1.4$$

$$70 (0.25)^{1.25} = p_2 (0.03)^{1.25}$$

$$p_2 = 991.1 \text{ kPa}$$

$$W = \frac{p_2 V_2 - p_1 V_1}{1 - 1.25}$$

$$1 - 1.25$$

$$W = -48.932 \text{ kJ}$$

$$Q = -48.932 \left(\frac{1.4 - 1.25}{1.4 - 1} \right)$$

$$Q = -18.3495 \text{ kJ}$$